Graph Machine Learning

Centrality Measures

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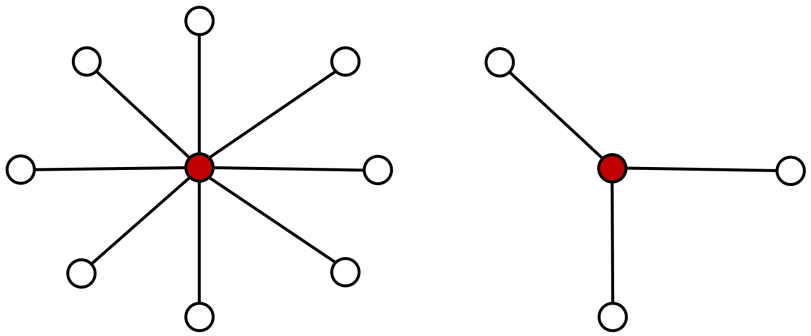
Centrality Measures

Centrality Measures

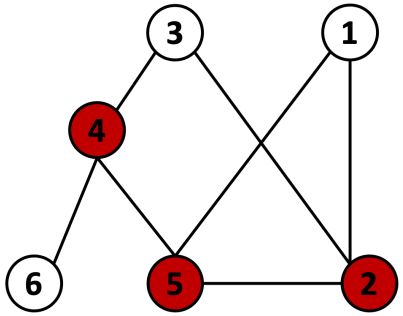
- ☞ Centrality indices are answers to the question **"What characterizes an important vertex?"** Need to define a real-valued function on the vertices of a graph, to provide a ranking which identifies **the most important nodes**.
- ☞ Centrality has a wide number of meanings, leading to different definitions of centrality:
  $\longrightarrow$ cohesiveness, ability to transfer information across the network, to affect/influence the others, to control information flow, ...
- ☞ Many centrality measures count the number of **paths (or walks)** through a given vertex; the measures differ in how the relevant walks are defined and counted: from walks of length one (degree centrality) to infinite walks (eigenvector centrality).
- ☞ Other centrality measures, such as betweenness centrality focus not just on overall connectedness but occupying **positions that are pivotal** to the network's connectivity.
  $\longrightarrow$ Influencers in social networks, vehicles for disease spreading, hubs in road networks, key infrastructures on the Internet, ...

Degree Centrality

- More neighbors (connections) a node has high potential communication activity, more important it is!



- The matrix form: **c** = **A1** where is the all one vector



$$\begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 2 \\ 3 \\ 3 \\ 1 \end{pmatrix}$$

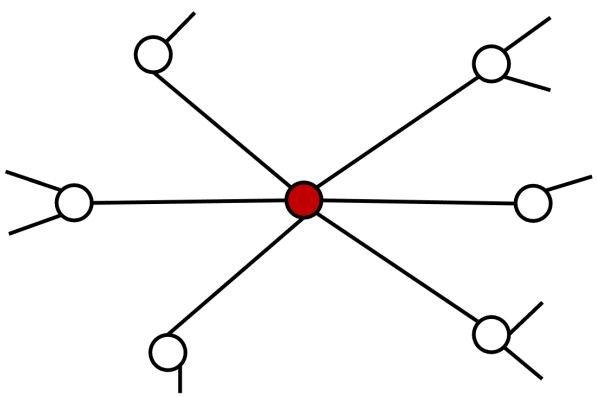
- Very likely that more than one vertex has the same degree $\rightarrow$ difficult to uniquely rank the vertices

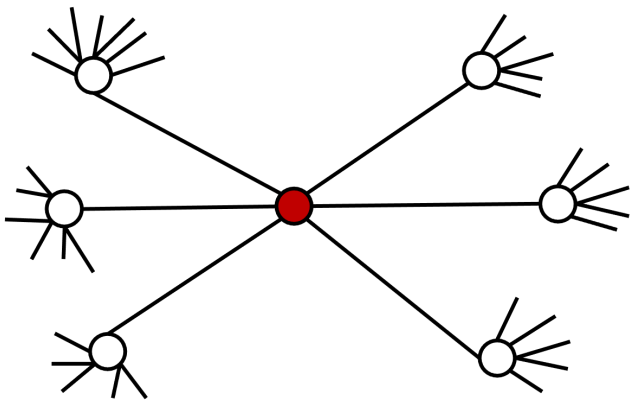
Neighborhood connectivity

☞ The average degree of v's neighbors, more this average is high for a node more it is important

$$c_v = \frac{\sum_{u \in \mathcal{N}_v} d_u}{d_v}$$

$$\mathbf{c} = \mathbf{D}^{-1}\mathbf{A}\mathbf{1}$$





Eigenvector centrality

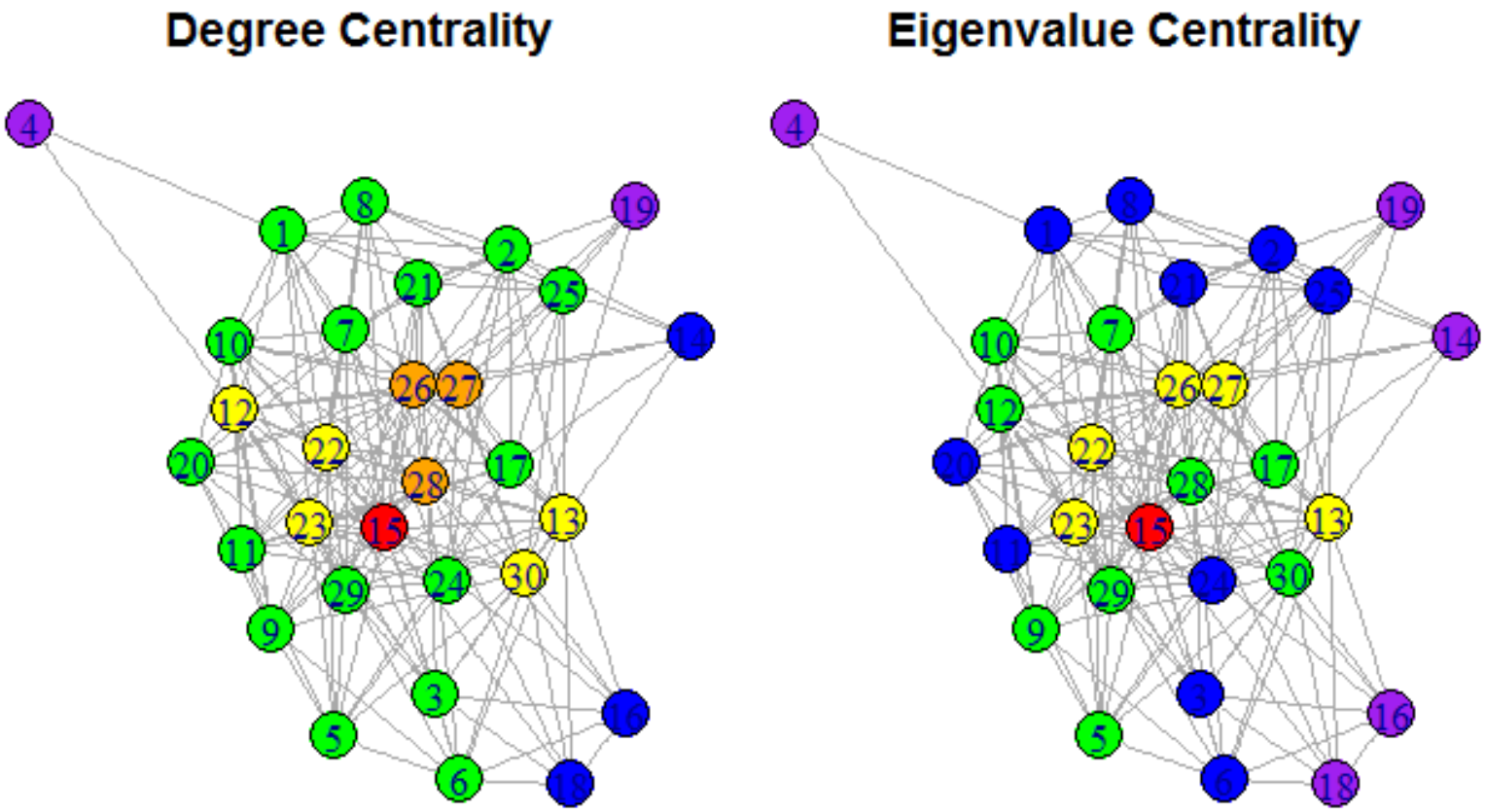
- A natural extension of degree centrality is eigenvector centrality
  ✓ Eigenvector centrality measures a node's importance while giving consideration to the importance of its neighbors. A high eigenvector score means that a node is connected to many nodes which themselves have high scores.
- The eigenvector centrality c_{v_i} of v_i is the sum of the centralities of its neighbors: $c_{v_i} = \frac{1}{\lambda} \sum_{v_j \in \mathcal{N}_i} \mathbf{A}_{ij}\, c_{v_j}$

$$\mathbf{A}\mathbf{c} = \lambda \mathbf{c}$$

- Introduced by the sociologist Philip Bonacich (1972) suggested that the eigenvector of the largest eigenvalue of an adjacency matrix could make a good network centrality measure
- The eigenvector $\mathbf{c}$ must be non-negative, according to P-F theorem only the largest λ_{max}, the dominant value, results in the desired centrality measure.
- Example: $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$; $|\mathbf{A} - \lambda \mathbf{I}| = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 3 - 4\lambda + \lambda^2 = 0$

  $\mathbf{E}_{\lambda=3} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

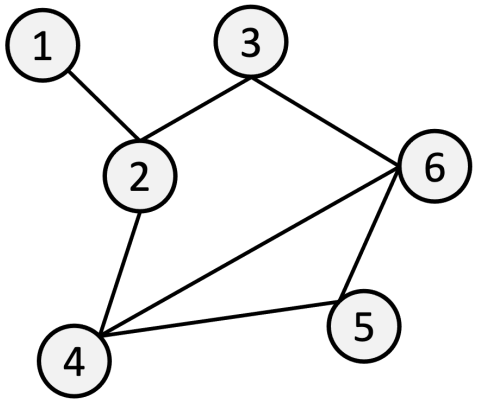
Eigenvector Centrality *versus* Degree centrality



from "purple","blue","green","yellow","orange","red"

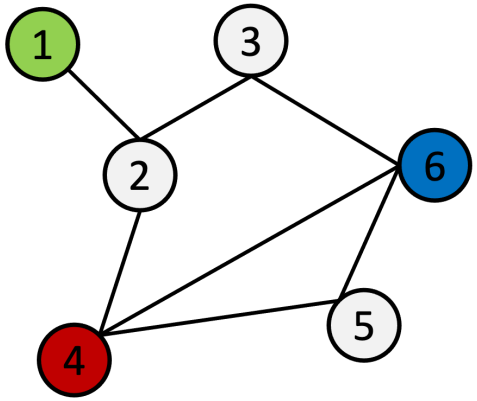
Eigenvector centrality: Power method

☞ Power method $\mathbf{c}^{(k+1)} = \mathbf{A}\mathbf{c}^{(k)}$. Initially the centralilty of each node is 1.



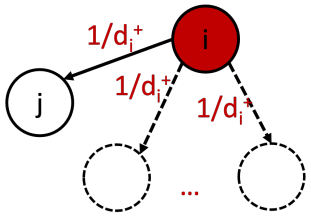
Eigenvector centrality

☞ Power method $\mathbf{c}^{(k+1)} = \mathbf{A}\mathbf{c}^{(k)}$
4, 6, 2, 5, 3, 1



PageRank Centrality

- Google's PageRank is a variant of the eigenvector centrality. PageRank uses in-degree to award one centrality point (popularity) for every link a node receives.



$$\mathbf{P}_{ij} = \begin{cases} \frac{1}{d_i^+}, & \text{if } v_j \in \mathcal{N}_i^+ \\ 0, & \text{otherwise} \end{cases};$$

$\frac{1}{d_i^+}$ represents the probability for a surfer to jump randomly from page v_i to a page v_j (independent events). $\mathbf{P}$ is a row stochastic $\sum_j \mathbf{P}_{ij} = 1$.

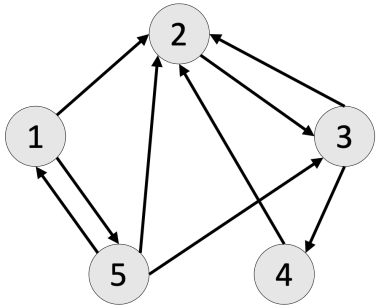
- The matrix form is defined as follows:

$$\lambda \mathbf{c} = \mathbf{c}\mathbf{P} \text{ with } \lambda = 1$$

$$\mathbf{c} = (\mathbf{D}^+)^{-1}\mathbf{A}\mathbf{1}$$

PageRank Centrality

☞ We use power iteration method: $\mathbf{c}^{(k+1)} = \mathbf{c}^{(k)}\mathbf{P}$; $\mathbf{c}^{(k)} = \mathbf{c}\mathbf{P}^k$ with $\mathbf{c} = \begin{bmatrix} \frac{1}{n} & \frac{1}{n} & \dots & \frac{1}{n} \end{bmatrix}$



$$\mathbf{P} = \begin{bmatrix} 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 \end{bmatrix} \text{ and } \mathbf{c} = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \end{bmatrix}$$

$$\mathbf{c}^{10} = \begin{bmatrix} 0 & 0.4 & 0.4 & 0.2 & 0 \end{bmatrix}$$

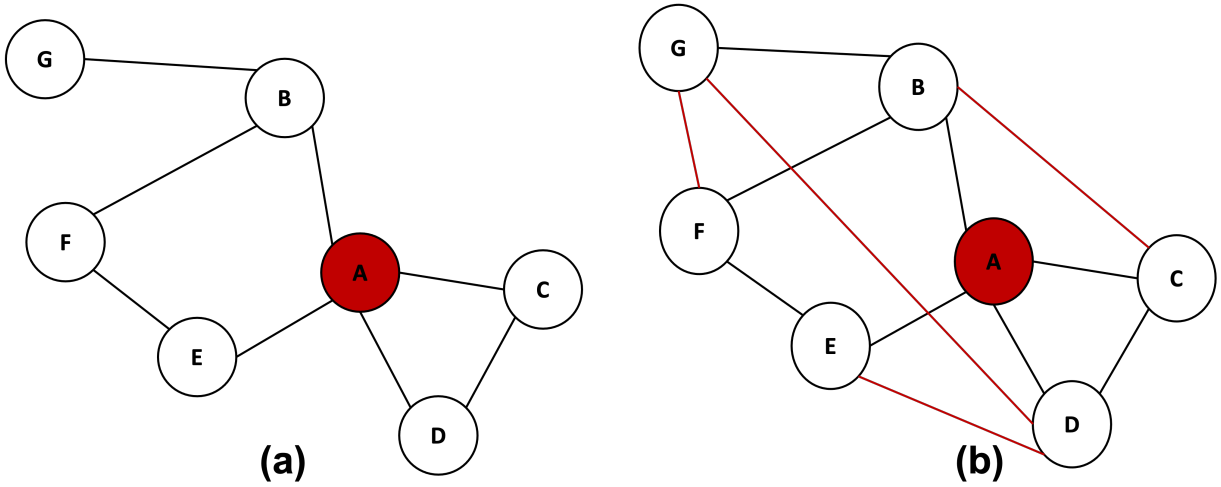
PageRank Centrality

☞ PageRank algorithm holds that a surfer who is randomly clicking on links will eventually stop clicking. The probability, at any step, that the surfer will continue is a damping factor α Damping factor d

$$\mathbf{c} = \begin{bmatrix} \frac{(1-\alpha)}{n} & \frac{(1-\alpha)}{n} & \dots & \frac{(1-\alpha)}{n} \end{bmatrix} + \alpha \mathbf{c}\mathbf{P}$$

Clustering Coefficient Centrality

- Triadic Closure: *If two people in a social network have a friend in common, then there is an increased likelihood that they will become friends themselves*.
- The probability that two randomly selected friends of node A are friends with each other:
  - ✓ Figure (a) = $\frac{1}{6}$ because there is only the single (C, D) edge among the six pairs of (B, C), (B, D), (B, E), (C, D), (C, E), and (D, E) increased to $\frac{1}{2}$ in Figure (b) three edges (B, C), (C, D), and(D, E) among the same six pairs.

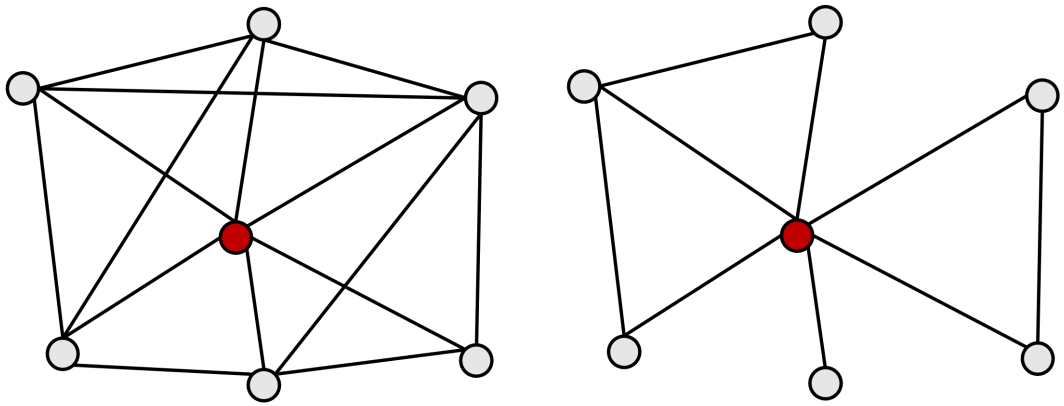


Clustering Coefficient Centrality

☞ Clustering Coefficient cc_v captures how the neighbors of v are linked to each other.

$$c_v = \frac{\left|\{(v,u),(u,w),(w,v) \in E\}\right|}{\frac{1}{2}d_v(d_v-1)}$$ where $\frac{1}{2}d_v(d_v-1)$ is the total number of links between neighbors

$\left|\{(v,u),(u,w),(w,v) \in E\}\right|$ is the number of all triangles involving v and v's neighbors



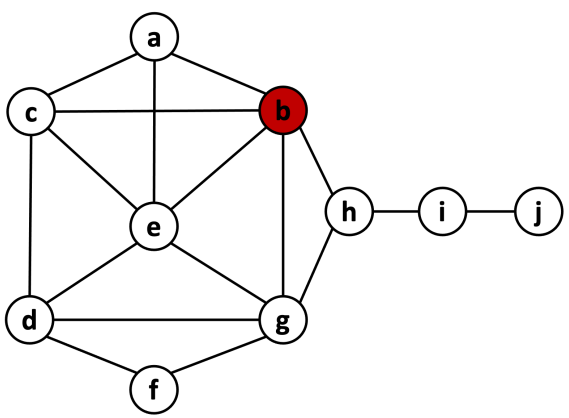
☞ The more densely interconnected v's neighborhood, the higher is its clustering coefficient. $c_v = 0$ v's neighborhood are not connected, $c_v = 1$ v's neighborhood are all connected.

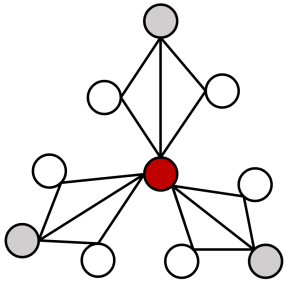
Closeness centrality

- Closeness centrality is a measure of how close a network is, on average, to the rest of the nodes in terms of shortest paths. It measures the average distance between a node v and all other nodes in the network. Thus, the more central a node is, the closer it is to all other nodes.

$$c_V = \frac{1}{\sum_{r \neq v} dist(v, r)}$$

it can be normalised by $\frac{\sum_r dist(v,r)}{|V|-1}$





- Harmonic centrality: $c_V = \sum_{r \neq v} \frac{1}{dist(v, r)}$. The inverse distance to an unreachable vertex is considered to be zero $\frac{1}{dist(v,r)} = 0$

Betweenness centrality

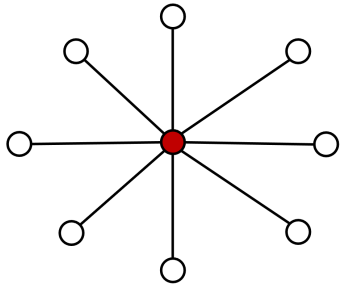
- A family of betweenness measures are defined to capture a node's importance as a conduit of information flow in a network
- Wide applications in network theory: in a telecommunications network, a node with higher betweenness centrality would have more control over the network, because more information will pass through that node.
- The most well known measures the number of times a node is on a shortest path between two nodes.

$$c_v = \sum_{s \neq v \neq t} \frac{\sigma_{s,t}(v)}{\sigma_{s,t}}$$

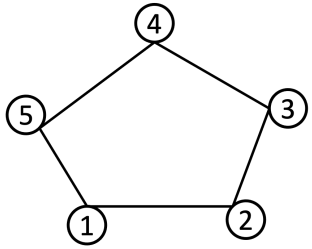
  where $\sigma_{s,t}$: number of shortest paths between source node s and target node t, while $\sigma_{s,t}(v)$: number of shortest paths between source s and target t nodes that pass through v node.
- It may be normalised by the number of ordered pairs of vertices not including v (combination of 2 over $(n-1)$).
  - ✓ for directed graphs $c_v = \frac{1}{(n-1)(n-2)} \sum_{s \neq v \neq t} \frac{\sigma_{s,t}(v)}{\sigma_{s,t}}$
  - ✓ for undirected graphs $c_v = \frac{2}{(n-1)(n-2)} \sum_{s \neq v \neq t} \frac{\sigma_{s,t}(v)}{\sigma_{s,t}}$

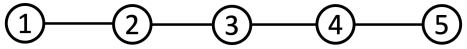
Betweenness centrality

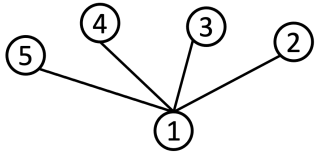
- ☞ undirected star graph, the center vertex has a betweenness of 1, while the leaves have a betweenness of 0.

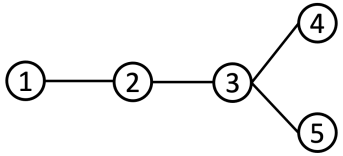


- ☞ What about the following graphs?



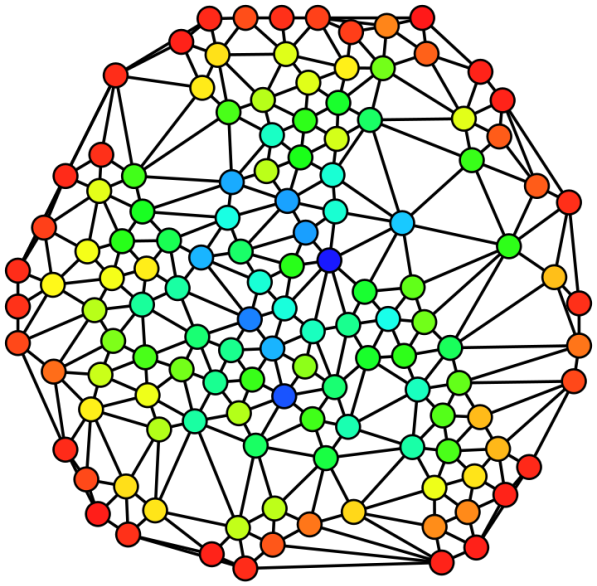






Betweenness centrality

- This graph shows the node betweenness, from red nodes to blue nodes = max,



- Betweeness assumes that information flow is through the shortest path. In a transport, the traffic will likely go through alternatives paths. Also in the case of rumours or infection.
- Katz centrality takes into account all paths of length k
- PageRank centrality takes into account the most probable walks.